**Example 5.24**

Calculate the mode from the following data

| Days of Confinement | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|
| Number of patients | 4 | 6 | 7 | 5 | 3 |

***Solution:***

Here, 7 is the maximum frequency, hence the value of x corresponding to 7 is 8. Therefore 8 is the mode.

## (b) Mode for Continuous data:

The mode or modal value of the distribution is that value of the variate for which the frequency is maximum. It is the value around which the items or observations tend to be most heavily concentrated. The mode is computed by the formula.

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times c$$

Modal class is the class which has maximum frequency.

$f_1$ = frequency of the modal class

$f_0$ = frequency of the class preceding the modal class

$f_2$ = frequency of the class succeeding the modal class

$c$ = width of the class limits

**Remarks**

(i) If $(2f_1\text{-}f_0\text{-}f_2)$ comes out to be zero, then mode is obtained by the following formula taking absolute differences $M_0 = l + \left( \frac{(f_1 - f_0)}{|f_1 - f_0| + |f_1 - f_2|} \times C \right)$

(ii) If mode lies in the first class interval, then $f_0$ is taken as zero.

(iii) The computation of mode poses problem when the modal value lies in the open-ended class.

**Example 5.25**

The following data relates to the daily income of families in an urban area. Find the modal income of the families.



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